Multi-Cloud Asset Cost Optimization Portal

Problem Statement #45

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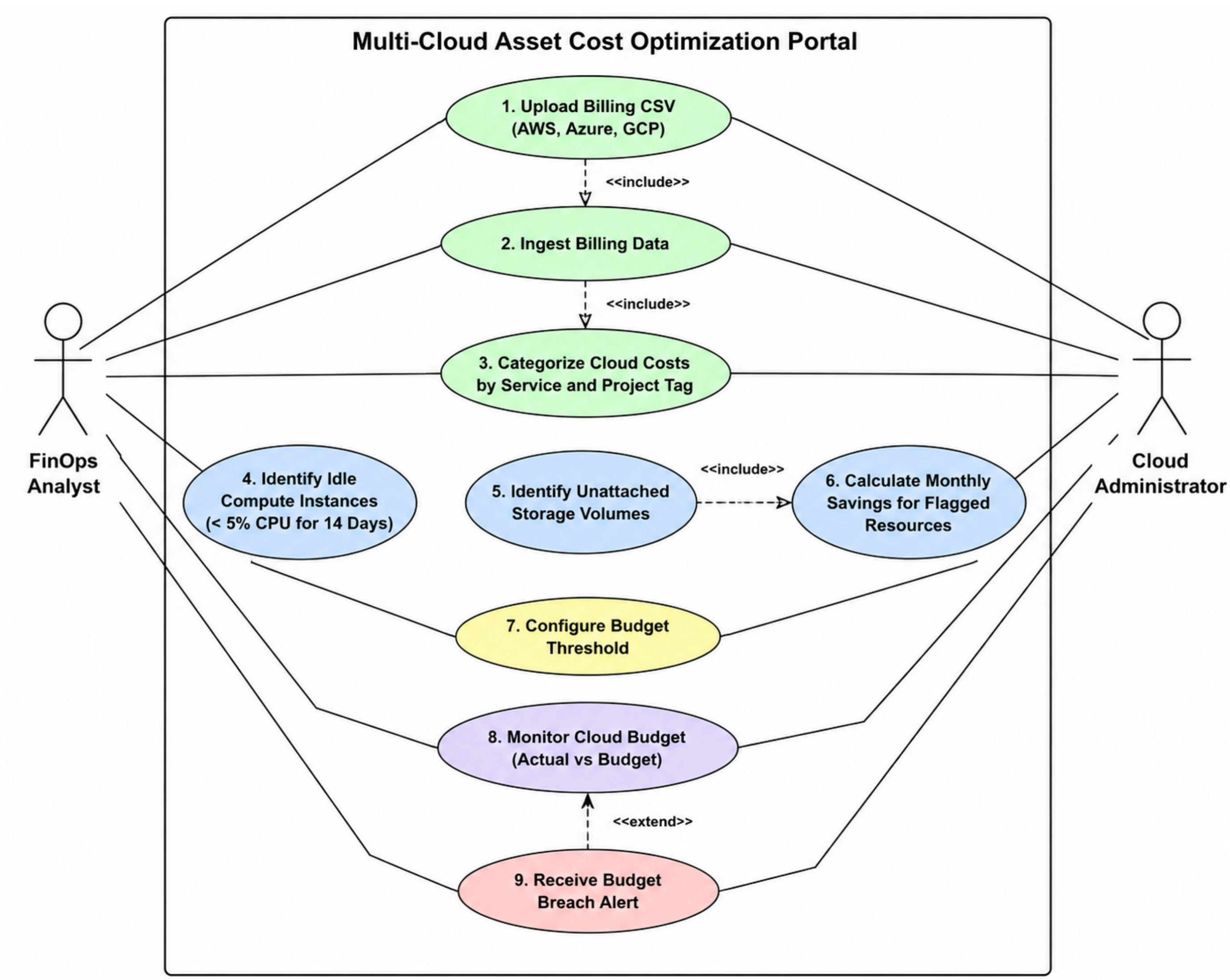
1. Functional Requirements (FR)

ID	Requirement	Action	Association
FR-001	The system shall ingest cloud billing CSV files from AWS, Azure, and GCP.	Upload / Ingest Billing Data	FinOps Analyst, Cloud Administrator
FR-002	The system shall categorize cloud costs by service and project tag.	Categorize Cloud Costs	FinOps Analyst
FR-003	The system shall identify compute instances with less than 5% CPU utilization over 14 days.	Identify Idle Compute Instances	FinOps Analyst, Cloud Administrator
FR-004	The system shall identify unattached storage volumes and provide a monthly savings estimate.	Identify Unattached Storage / Calculate Savings	FinOps Analyst, Cloud Administrator
FR-005	The system shall generate an alert when cloud spending exceeds the configured budget threshold.	Monitor Budget / Generate Alert	FinOps Analyst, Cloud Administrator

2. Non-Functional Requirements (NFR)

ID	Requirement	Action	Association
NFR-001	The portal shall parse multi-gigabyte billing files and update cost dashboards in under 2 minutes.	Process Billing Data / Update Dashboard	System
NFR-002	The portal shall protect billing information and allow access only to authorized users.	Authenticate / Authorize Access	FinOps Analyst, Cloud Administrator

3. USE CASE Diagram



4. USE CASE FLOW Specification

1. Preconditions

1. The user is authorized to access the portal.
2. Cloud billing data has been uploaded and ingested.
3. Compute utilization data for the required 14-day period is available.
4. The system is operational.

2. Postconditions

Success:

- Idle compute instances are identified and flagged.
- The flagged instances are displayed on the dashboard.
- A monthly savings estimate is provided for the flagged resources.

Failure:

- Instances without sufficient utilization data are not incorrectly classified as idle.
- The system informs the user that the analysis could not be completed for those instances.

3. Main Success Scenario

5. The FinOps Analyst logs into the Multi-Cloud Asset Cost Optimization Portal.
6. The analyst selects "Identify Idle Compute Instances."
7. The system retrieves compute instance utilization data.
8. The system examines CPU utilization for each compute instance over the previous 14 days.
9. The system compares the CPU utilization against the 5% threshold.
10. The system identifies instances with CPU utilization below 5%.
11. The system flags those instances as idle.
12. The system calculates the estimated monthly savings associated with the flagged instances.
13. The system displays the flagged instances and their estimated savings on the dashboard.
14. The FinOps Analyst reviews the results.

4. Alternate Flow

AF-01 – Insufficient Utilization Data

15. At Step 4, the system determines that an instance does not have sufficient CPU utilization data for the complete 14-day period.
16. The system does not classify that instance as idle.
17. The system marks the instance as "Insufficient Data."
18. The system continues analyzing other compute instances for which sufficient data is available.
19. The system displays the instances with insufficient data separately.
20. The FinOps Analyst can review the remaining valid results.

5. Acceptance Criteria

- An instance with less than 5% CPU utilization over 14 days must be flagged as idle.
- An active instance must not be incorrectly classified as idle.
- The flagged resource must have a corresponding monthly savings estimate.